

FPGA ASSESSMENT 1

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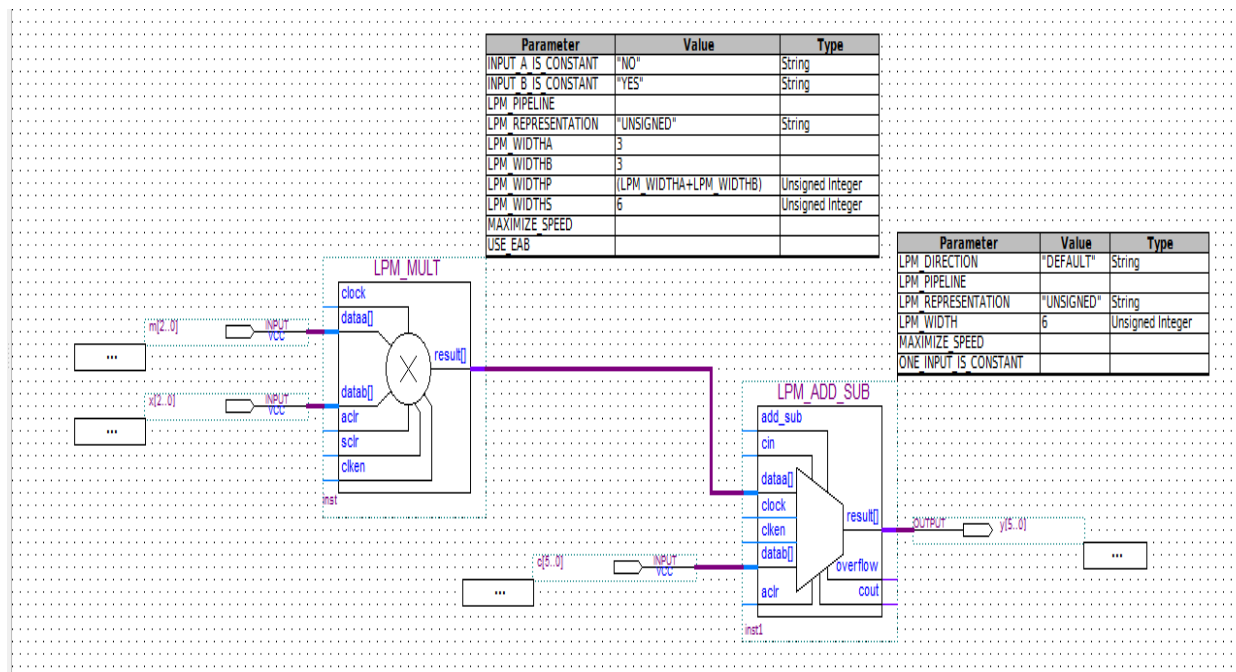
EXPERIMENT-1

DESIGN AND IMPLEMENTATION OF ARITHMETIC EXPRESSION

OBJECTIVE

To design, simulate, and implement the arithmetic function $Y=4X+5$ using Quartus Prime and verify its operation on the Altera DE2-115 FPGA.

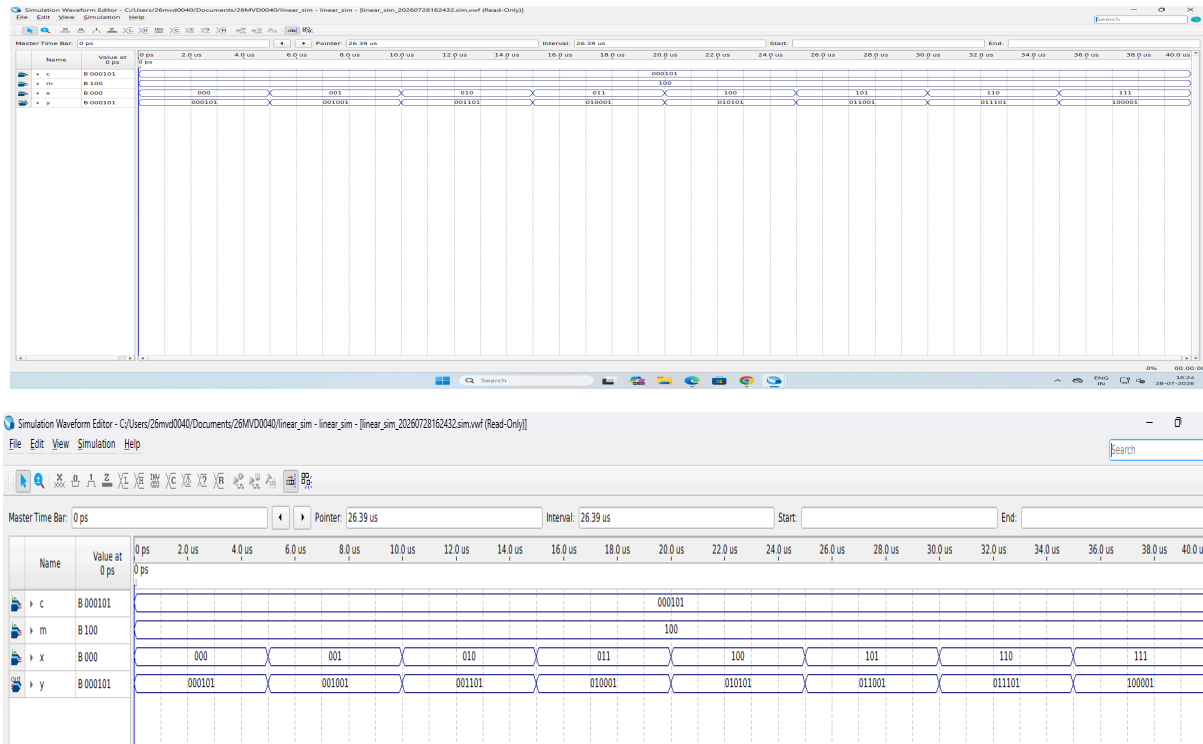
BLOCK DIAGRAM



The block diagram implements the arithmetic function $Y=4X+5$ using Quartus LPM (Library of Parameterized Modules). The LPM_MULT block multiplies the 3-bit input X by the constant value 4. The multiplication result is then applied to

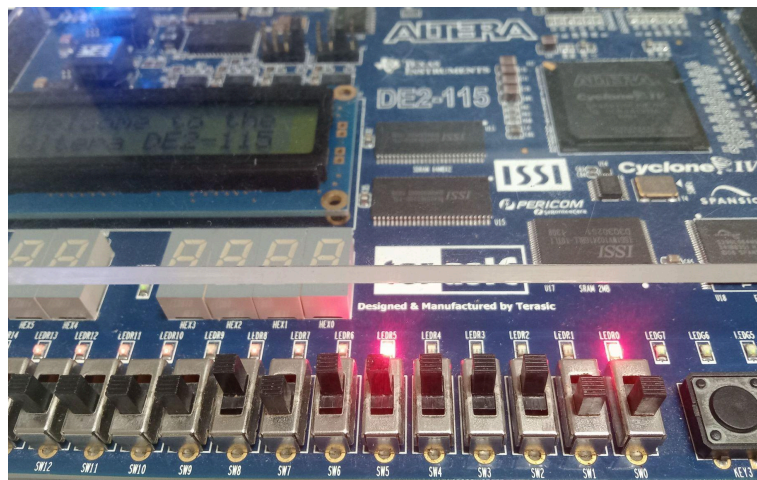
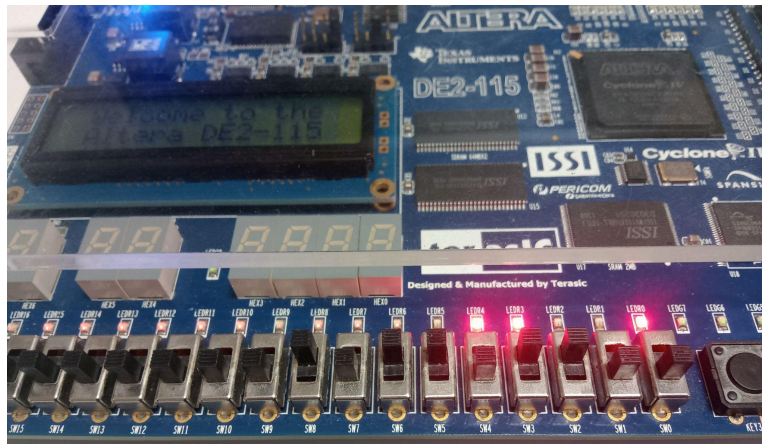
the LPM_ADD_SUB block, where the constant value 5 is added to produce the 6-bit output Y.

WAVEFORM



The simulation waveform verifies the correct operation of the designed circuit. As the input X changes from 000 to 111, the output Y follows the equation $Y=4X+5$. The observed outputs match the expected values, confirming the correctness of the design.

FPGA IMPLEMENTATION



The design was successfully compiled and programmed onto the Altera DE2-115 FPGA (Cyclone IV E) board. The input X was applied using the onboard slide switches, and the output Y was observed through the LEDs. The hardware output matched the simulation results.

For the tested inputs:

- $X = 101 \text{ (5)} \rightarrow Y = 25 \text{ (011001)}$

- $X = 111 (7) \rightarrow Y = 33 (100001)$

INFERENCE

The designed arithmetic circuit successfully implements the expression $Y=4X+5$. Both simulation and FPGA hardware verification produced identical outputs, demonstrating the correctness of the design and the functionality of the LPM multiplier and adder blocks.

CONCLUSION

The arithmetic function $Y=4X+5$ was successfully designed, simulated, and implemented on the Altera DE2-115 FPGA using Quartus Prime. The simulation and hardware results agreed with the theoretical values, confirming the successful realization of the design.